Taller Formativo #3

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Materia: Cálculo de una Variable - Práctico

(a) Hallar $f'(x)$ utilizando la definición de derivada, de: $f(x)=\log(\operatorname{sen}x)$, $x\in(0,\pi)$

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} \quad ; \quad f(x)=\log(\operatorname{sen}(x))$$

$$f'(x)=\lim_{h\to 0}\frac{\log(\operatorname{sen}x+h)-\log(\operatorname{sen}x)}{h} \Rightarrow f'(x)=\lim_{h\to 0}\frac{\log\left(\frac{\operatorname{sen}x+h}{\operatorname{sen}x}\right)}{h}=\lim_{h\to 0}\log\left(\frac{\operatorname{sen}x+h}{\operatorname{sen}x}\right)^{\frac{1}{h}}$$

$$f'(x)=\log\left(\lim_{h\to 0}\left(\frac{\operatorname{sen}x+h}{h}\right)^{\frac{1}{h}}\right)=\log\left(\lim_{h\to 0}\left(1+\left(\left(\frac{\operatorname{sen}x+h}{\operatorname{sen}x}\right)-1\right)\right)^{\frac{1}{h}}\right)$$

✓ Dado que se cumple la indeterminación 1^{∞}:

$$\left(\lim_{h\to 0}\frac{\operatorname{sen}(x+0)}{\operatorname{sen}(x)}\right)^{\lim_{h\to 0}\frac{1}{0}}=1^{\infty}$$

$$f'(x)=\log\left(\left(\underbrace{\lim_{h\to 0}\left(1+\left(\frac{\operatorname{sen}(x+h)-\operatorname{sen}x}{\operatorname{sen}x}\right)\right)^{\frac{\operatorname{sen}x}{\operatorname{sen}(x+h)-\operatorname{sen}x}}}\right)^{\frac{1}{h}\cdot\frac{\operatorname{sen}x+h}{\operatorname{sen}x}-1}\right)$$

Se puede utilizar el límite notable $\lim_{h\to 0}(1+h)^{\frac{1}{h}}=e$

$$f'(x)=\log\left(e^{\lim_{h\to 0}\frac{1}{h}\left(\frac{\operatorname{sen}(x+h)-\operatorname{sen}x}{\operatorname{sen}x}\right)}\right)$$

$$f'(x)=\log\left(e^{\lim_{h\to 0}\frac{\operatorname{sen}x\cosh+\operatorname{sen}h\cos x-\operatorname{sen}x}{h\cdot\operatorname{sen}x}}\right)=\log\left(e^{\lim_{h\to 0}\frac{-\operatorname{sen}x(1-\cos h)+(\cos x\operatorname{sen}h)}{h\operatorname{sen}x}}\right)$$

$$f'(x)=\log\left(e^{\lim_{h\to 0}-\frac{\cancel{\operatorname{sen}x}(1-\cos h)}{\cancel{\operatorname{sen}x}\cdot h}+\frac{\cos x\operatorname{sen}h}{\operatorname{sen}x\cdot h}}\right)=\log\left(e^{-\lim_{h\to 0}\frac{1-\cos h}{h}+\lim_{h\to 0}\frac{\operatorname{sen}h}{h}\cdot\lim_{h\to 0}\frac{\cos x}{\operatorname{sen}x}}\right)$$

$$\lim_{h\to 0}\frac{1-\cos h}{h}=0 \quad ; \quad \lim_{h\to 0}\frac{\operatorname{sen}h}{h}=1 \quad ; \quad \lim_{h\to 0}\frac{\cos x}{\operatorname{sen}x}=\frac{\cos x}{\operatorname{sen}x}$$

Aplicación de límites notables

$$f'(x)=\log\left(e^{\frac{\cos x}{\operatorname{sen}x}}\right) \Rightarrow f'(x)=\log\left(e^{\cot(x)}\right)$$

cambio de base al logaritmo: $f'(x)=\dfrac{\ln(e^{\cot x})}{\ln(10)}$

$$\boxed{f'(x)=\frac{\cot(x)}{\ln(10)}}$$

Notas

(b) Calcule la ecuación de la recta normal, para un punto en la abscisa:

$x = \frac{\pi}{4}$

1. Para $x = \pi/4 \Rightarrow f(\pi/4) = \log(\operatorname{sen}(\pi/4)) \Rightarrow f(\pi/4) = \log\left(\frac{\sqrt{2}}{2}\right)$

$\log\left(\frac{\sqrt{2}}{2}\right) = \log(2^{1/2}) - \log(2^{-1}) \Leftrightarrow \log\left(\frac{1}{\sqrt{2}}\right) = \log(1) - \log(2^{\frac{1}{2}})$

$\boxed{f(\pi/4) = -\frac{1}{2}\log(2)}$; $P(\pi/2, -\frac{1}{2}\log(2))$

2. para $x = \pi/4 \Rightarrow f'(\pi/4) = \frac{\cot(\pi/4)}{\ln(10)} \Rightarrow f'(\pi/4) = \frac{1}{\ln(10)}$

$\boxed{m_{tg} = \frac{1}{\ln(10)}} \Rightarrow$ dado que $m_{tg} m_n = -1 \Rightarrow \boxed{m_n = -\ln(10)}$

3. Ecuación de la recta normal: $y - y_0 = m_n(x - x_0)$

$P(\pi/4, -\frac{1}{2}\log(2)) \Rightarrow y - (-1/2\log(2)) = -\ln(10)(x - \frac{\pi}{4})$

$\boxed{y + \frac{1}{2}\log(2) = -\ln(10)(x - \pi/4)}$ //

$\boxed{L_n: \ x\ln(10) + y + \left(\frac{1}{2}\log(2) - \frac{\pi}{4}\ln(10)\right) = 0}$ //